Following standard terminology from quantum optics, we refer to all such combinations of $\hat{x}$ and $\hat{p}$ as *quadrature operators*. The rotated quadratures obey the same commutation relations as $\hat{x}$ and $\hat{p}$,

$$[\hat{x}(\theta), \hat{p}(\theta)] = i. \tag{3.55}$$

In addition to rotations, translations of the quadratures are also important. These are realised by the *displacement operator*, defined as

$$\hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}. \tag{3.56}$$

Since the exponent is anti-Hermitian, the displacement operator is unitary. We have $\hat{D}^\dagger(\alpha) = \hat{D}(-\alpha)$. Furthermore, using the Campbell identity again[3]

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha. \tag{3.57}$$

From here, we can see that the displacement shifts $\hat{x}$ and $\hat{p}$

$$\hat{D}^\dagger(\alpha)\hat{x}\hat{D}(\alpha) = \hat{x} + \sqrt{2}\,\mathrm{Re}(\alpha), \tag{3.58}$$

$$\hat{D}^\dagger(\alpha)\hat{p}\hat{D}(\alpha) = \hat{p} + \sqrt{2}\,\mathrm{Im}(\alpha). \tag{3.59}$$

This means that for any (normalised) pure state $|\psi\rangle$, a displacement $|\psi\rangle \to \hat{D}(\alpha)|\psi\rangle$ shifts the expectation values of $\hat{x}$ and $\hat{p}$ by $\sqrt{2}\,\mathrm{Re}(\alpha)$ and $\sqrt{2}\,\mathrm{Im}(\alpha)$, respectively. The variances, on the other hand, do not change (see Prob. 3.3). The operator $\hat{D}(\alpha)$ thus shifts the mean of states but leaves their noise properties unchanged (at least in $\hat{x}$ and $\hat{p}$). In this sense, it is a displacement. Below, we will also see that this corresponds to translations in phase space.

3.1.5 Coherent states

In addition to the energy eigenstates, there are of course many other interesting classes of quantum states of harmonical oscillators. One particularly important class is that of so-called *coherent states*. Coherent states are ubiquitous in quantum optical experiments and applications because they provide an accurate description of the state of light produced by lasers. Hence, they are among the easiest states to produce in practice. At the same time, they are in a sense the 'most classical' states of the quantum harmonic oscillator.

Formally, the coherent states are the eigenstates of the annihilation operator. For every complex number α, there exists a state $|\alpha\rangle$ such that

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle. \tag{3.60}$$

That is, $|\alpha\rangle$ is a left-eigenstate of $\hat{a}$. It is also a right-eigenstate of $\hat{a}^\dagger$ with eigenvalue α^*, i.e. $\langle\alpha|\hat{a}^\dagger = \langle\alpha|\alpha^*$. We take $|\alpha\rangle$ to be normalised. The mean number of excitations is seen to be

$$\langle\alpha|\hat{n}|\alpha\rangle = |\alpha|^2. \tag{3.61}$$

[3] For operators $\hat{A}$ and $\hat{B}$ such that $\hat{A}$ commutes with the commutator $[\hat{A}, \hat{B}]$, the Campbell identity simplifies to $e^{\hat{A}}\hat{B}e^{-\hat{A}} = B + [\hat{A}, \hat{B}]$. We can apply this result here with $\hat{A} = -(\alpha\hat{a}^\dagger - \alpha^*\hat{a})$ and $\hat{B} = \hat{a}$.